

NMAP & FIREWALL LAB CHEATSHEET

1. BASIC FIREWALL COMMANDS (IPTABLES)

`iptables -L FORWARD -v -n --line-numbers`

List FORWARD rules with details & line numbers

`iptables -F FORWARD`

Flush (delete) all rules in FORWARD chain

`iptables -P FORWARD ACCEPT`

Set default FORWARD policy to ACCEPT (allow all)

`iptables -P FORWARD DROP`

Set default FORWARD policy to DROP (block all)

2. FIREWALL SCENARIO CONFIGURATIONS

`iptables -A FORWARD -p tcp --dport 80 -j DROP`

Silently drop TCP port 80 traffic (causes timeout)

`iptables -A FORWARD -p tcp --dport 22 -j REJECT`

Reject SSH port 22 with ICMP port-unreachable

`iptables -A FORWARD -p tcp -d <target> --dport 80 -j ACCEPT`

Stateless: Allow incoming TCP SYN to port 80

`iptables -A FORWARD -p tcp -s <target> -j ACCEPT`

Stateless: Allow return traffic from target host

`iptables -A FORWARD -m state --state ESTABLISHED -j ACCEPT`

Stateful: Allow replies to active connections

`iptables -A FORWARD -p tcp --dport 80 -m state --state NEW -j ACCEPT`

Stateful: Allow NEW SYN connections to port 80

`iptables -A FORWARD -p tcp --sport 53 -j ACCEPT`

Weak Rule: Naively trust source port 53 (DNS)

3. NMAP NETWORK SCANNING COMMANDS

`nmap -sT -p 22,80,443 <target>`

TCP Connect scan (completes 3-way handshake)

`nmap -sS -p 22,80,443 <target>`

TCP SYN stealth scan (half-open, root required)

`nmap -sU -p 53,67,161 <target>`

UDP scan (checks non-TCP service ports)

4. FIREWALL ENUMERATION & DIAGNOSTICS

`nmap -sA -p 22,80,443 <target>`

ACK scan (maps stateless vs stateful rules)

`nmap -sN -p 22,80,443 <target>`

NULL scan (no flags set; tests RFC 793 compliance)

`nmap -sF -p 22,80,443 <target>`

FIN scan (FIN flag set; probes open/closed response)

`nmap -sX -p 22,80,443 <target>`

Xmas scan (FIN, PSH, URG flags set; stealth probe)

`nmap -sS -p 22,80 --packet-trace <target>`

Trace raw packets to differentiate DROP vs REJECT

`tcpdump -i eth0 host <target> -n`

Capture network traffic to verify scan packets

5. FIREWALL EVASION & BYPASS TECHNIQUES

`nmap -sS -g 53 -p 22 <target>`

Source port spoofing (bypasses weak sport 53 rule)

`nmap -sS -f --mtu 16 -p 80 <target>`

Fragment IP packets into 16-byte MTU to evade NIDS

`nmap -sS -D 10.10.10.5,ME <target>`

Decoy scan (obscures real IP among fake decoys)